

A CONDITIONAL SIX-QUANTIFIER DEFINITION OF $\mathbb{Z}$ INSIDE $\mathbb{Q}$: MODEL-FIELD ASSEMBLY, ITS WALLS, AND A MACHINE-VERIFIED FINITE PROGRAM

H10Q PROJECT

ABSTRACT. We describe a constructive route to a universal definition of $\mathbb{Z}$ inside $\mathbb{Q}$ using six quantifiers, conditional on one explicit hypothesis H about the solubility of a family of tied conics. Every finitary component of the construction is machine-verified on a grid of 353 cells (w, z) with w an odd prime below 100: all 353 cells carry a verified witness, and all 293 aligned classes carry a certified emergent-free member. Two structural theorems delimit the search space: a generalized wall showing that every b coprime to the cell data fails on every branch — valid for composite squarefree A as well as prime A — and a characterization theorem stating that an aligned class exists for a cell exactly when the numerator of z has a prime divisor congruent to 1 modulo 4. The residual obstruction is analysed by an exact p -adic sieve model whose matched predictions agree with the observed obstruction at member level on an identical 8,760-member sample (marginal ratio 0.9993; hit-conditioned union residual -2.9 ; marginal-preserving permutation null $z = -0.09$). The analytic input is then identified exactly: we prove that *Schinzel’s Hypothesis H for the pair $\{q_1 + Nt, G(t)\}$ implies the per-class hypothesis*, indeed infinitely often, so the conditional statement rests on a named classical conjecture rather than a bespoke one; the Schinzel hypotheses themselves are verified exactly on all 293 canonical pairs. The verified computational horizon extends off-grid to fifteen further cells, including three that are provably obstructed for the canonical protocol — the obstruction being $w \equiv 11, 19 \pmod{20}$, a reciprocity phenomenon we prove and then break with a non-canonical class family. We claim no unconditional record: the published record remains the refereed ten-quantifier definition, and our statement is conditional on Schinzel’s Hypothesis H together with the finite class-existence side.

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1. INTRODUCTION

Hilbert's Tenth Problem over $\mathbb{Q}$ asks for an algorithm which, given a finite system of polynomials with rational coefficients, decides whether the system has a common rational zero. The problem remains open in both directions. One standard route to undecidability would be an existential definition of $\mathbb{Z}$ in $\mathbb{Q}$, since such a definition would transport the undecidability over $\mathbb{Z}$ to rational points. The converse is not known: failure of that route would not imply decidability over $\mathbb{Q}$.

There is already a sharp first obstruction. Cornelissen–Zahidi proved that $\mathbb{Z}$ cannot be the projection of the rational points of a curve over $\mathbb{Q}$; in particular, no polynomial with one existential witness defines $\mathbb{Z}$ in $\mathbb{Q}$ [4]. Thus two witnesses, equivalently a surface as witness variety, form the first open existential case. This is also exactly where the available topological and thin-set lower-bound methods stop. A different elliptic-curve route has its own cliff. If only finitely many primes are withheld from an S -integer ring, the indices n for which $x(nP)$ is integral are a finite union of congruence classes, relative to the chosen integral Weierstrass model. When nonempty this set has positive density and is archimedeanly equidistributed on a real component, rather than discrete. Consequently the prime-index mechanism used in elliptic Diophantine models degenerates at the cofinite- S boundary; this statement does not rule out unrelated predicates.

Koenigsmann changed the quantitative landscape by proving that $\mathbb{Z}$ is purely universally definable in $\mathbb{Q}$ [9]. Complementation turns this into the study of the existential rank $\text{rk}_{\mathbb{Q}}^{\exists}(\mathbb{Q} \setminus \mathbb{Z})$. Daans–Dittmann–Fehm then supplied the geometric interpretation: for a non-quantifier-free Diophantine set D over $\mathbb{Q}$,

$$\text{rk}_{\mathbb{Q}}^{\exists}(D) = \text{efd}_{\mathbb{Q}}(D) + 1,$$

and $\text{efd}_{\mathbb{Q}}(D) \leq d$ precisely when D is the image of a morphism all of whose fibres have dimension at most d [6]. The quantifier race is therefore a fibre-dimension problem. In particular, a six-universal-quantifier definition would amount to the bound $\text{efd}_{\mathbb{Q}}(\mathbb{Q} \setminus \mathbb{Z}) \leq 5$, not merely to more efficient logical notation.

1.1. The record chain and its audit. The refereed benchmark is Daans’s ten-quantifier universal definition [5]. Our source-level audit reconstructs its budget before applying any fusion: two quaternion parameters, three witnesses for the auxiliary Φ condition, one splitting witness, and two three-variable trace-norm certificates give twelve variables. Two uses of the intersection theorem of Daans–Dittmann–Fehm save one coordinate each, giving ten; pairwise reassociation cannot improve that proof shape. The same audit isolates the real places to seek an improvement: stronger simultaneous fusion, a cheaper trace-norm primitive, or a joint treatment of the nested valuation-ring conditions.

Sun subsequently announced an unrefereed seven-quantifier construction [13]. It retains Daans’s bridge but replaces the six-witness intersection block by a finite disjunction of three-witness quadrics, and then uses one further fusion. Our A2 audit checked the exact quaternion identities, the block’s soundness, the residue-field combinatorics, and explicit bridge certificates. It also traced the local–global completeness argument through parameterized Hensel lifting, weak and norm approximation, and Hasse–Minkowski. No substantive missing implication was found, but the posted preprint has stale cross-references at the decisive step, and its local–global chain was hand-traced rather than machine-verified. We therefore call seven an *unrefereed announcement*; ten remains the refereed record.

PROVED. The audit also corrects one ingredient of the unrefereed announcement [13]. Its finite-field selection lemma assumes a residue field larger than 25. Exhaustion first showed that the small fields do not fail; more conceptually, we proved the exact identity

$$N(\tau) = \frac{\#E_{\tau^{-2}}(\mathbb{F}_q)}{4}, \quad E_{\tau^{-2}} : y^2 = x(x-1)(x-\tau^{-2}).$$

Hasse’s bound makes this positive over every odd finite field admitting an admissible τ . Hence the threshold can be deleted: $\mathbb{F}_3$ is exceptional only because it has no $\tau \notin \{0, \pm 1\}$, while the preprint’s other, data-dependent exceptional places remain untouched.

1.2. Contributions and scope. The contributions of this paper are deliberately separated by status.

- (1) **CONDITIONAL — a six-quantifier architecture.** We tie a witness in Sun’s quadratic block to Daans’s congruence parameter by $a = 1 + 2s$. Soundness, denominator clearing, the exact target-place selector, and the global norm obstruction are proved; completeness is the explicitly named hypothesis H . If H holds, then $\mathbb{Q} \setminus \mathbb{Z}$ has existential rank at most six, $\mathbb{Z}$ has a six-quantifier universal definition, and the essential fibre dimension is at most five.
- (2) **PROVED — wall theorems.** We prove the two-adic source–target alignment wall, show that branch choice cannot remove the reciprocity obstruction, and prove the generalized wall: every admissible b whose odd support is coprime to the cell data fails. The only available door is controlled non-coprimality.
- (3) **PROVED — complete class characterization.** For the prime- b family, on every branch, an aligned class exists exactly when the numerator of z has a prime divisor congruent to 1 modulo 4. The necessity holds for every admissible rational a , not only when $1 + 4a^2$ is prime; the sufficiency is constructive.
- (4) **PROVED — exhaustive finite verification.** On the declared grid, every one of the 353 cells has a replayable soluble witness, and every one of the 293 aligned classes has a certified emergent-free member. These are bounded, per-row theorems under the proven-primality engine, not an extrapolation to all primes.
- (5) **EVIDENCE — a mechanistic sieve with matched validation.** Exact root lifting turns each recurring small-prime obstruction into an odd-valuation p -adic mass. On the identical 8,760-member sample, the matched marginal ratio is 0.9993, the hit-conditioned union residual is -2.9 , and a marginal-preserving permutation gives $z = -0.09$. These

data validate the small-layer mechanism in the measured window; they do not supply a tail bound or a uniform density theorem.

- (6) **PROVED — an audited Schinzel reduction.** For all 293 canonical linear/octic pairs, exact certificates verify irreducibility of the degree-8 factor, fixed-divisor and pair-product gcd conditions, and every local condition. The linear Dirichlet side is a theorem. The simultaneous prime-value assertion for the octic factor is not proved; it is precisely the remaining Schinzel input.
- (7) **PROVED — off-grid horizon extension.** The same certificate machinery closes and replays the four cells at $u = -1$, with $w = 101$, $w = 103$, $w = 107$, and $w = 109$. They lie beyond the verified grid horizon $w \leq 97$. This is an extension by four exact instances, not a uniform all- w theorem.

What is not claimed. We claim *no new unconditional quantifier record*: Daans's ten remains the refereed record, and Sun's seven remains an unrefereed announcement. The project hypothesis H is open. In particular, prime values of the irreducible degree-8 factor in the prescribed progressions are open; the closest unconditional square-class results have degree at most two, while the degree-three result is conditional on the elliptic-curve Parity Conjecture [10]. The finite grid, the matched sieve statistics, and the clean local audit do not prove that missing prime-value statement.

2. MODEL-FIELD ARCHITECTURE: SELECTION, WITNESS TYING, AND THE OPEN NORM PROBLEM

2.1. The finite-field selector is exact.

Theorem 2.1 (PROVED: L4, the small-field exhaustion). *Let q be an odd prime power with $q \leq 25$, let χ be the quadratic character of $\mathbb{F}_q$ with $\chi(0) = 0$, and let $\tau \in \mathbb{F}_q \setminus \{0, \pm 1\}$. There is an $a \in \mathbb{F}_q$ such that*

$$\chi(1 + 4a^2) = -1, \quad \chi(-(1 + 4a^2)(1 - \tau^2(1 + 4a^2))) = 1.$$

For $q = 3$ the assertion is vacuous, because there is no admissible τ . For $q \in \{5, 7, 9, 11, 13\}$ the minimum number of choices is 2, and for $q \in \{17, 19, 23, 25\}$ it is 4.

Thus the constant 25 in the unrefereed character-sum proof of [13] is not a genuine small-field obstruction: among odd residue fields, only the field of order 3 remains, and there the issue is admissibility rather than failure of the selector.

Theorem 2.2 (PROVED: L5, the Legendre identity). *For every odd prime power q , every $\tau \in \mathbb{F}_q \setminus \{0, \pm 1\}$, and $\lambda = \tau^{-2}$, let $E_\lambda : y^2 = x(x - 1)(x - \lambda)$. Then*

$$N(\tau) := \#\{a \in \mathbb{F}_q : \chi(1 + 4a^2) = -1, \chi(-(1 + 4a^2)(1 - \tau^2(1 + 4a^2))) = 1\} = \frac{\#E_\lambda(\mathbb{F}_q)}{4}.$$

Consequently

$$\#E_\lambda(\mathbb{F}_q) \geq (\sqrt{q} - 1)^2 > 0, \quad N(\tau) \geq (\sqrt{q} - 1)^2/4 > 0,$$

so the hypothesis $|k| > 25$ can be deleted from the lemma.

The proof is an exact point count, not an asymptotic substitution. With $u_a = 1 + 4a^2$ and $w_a = -u_a(1 - \tau^2 u_a)$, expand the corrected indicator; the hyperbola identity $\sum_b \chi(b^2 + c) = -1$ for $c \neq 0$ evaluates the quadratic pieces. Fibre-counting the map $a \mapsto u_a$ then gives

$$4N = q + 1 + \sum_u \chi(u(u - 1)(u - \lambda)) = \#E_\lambda(\mathbb{F}_q).$$

This also explains divisibility by 4: the Legendre curve has its full rational 2-torsion.

2.2. The tied model and its six-variable bookkeeping. Set $K = \mathbb{Q}$, $S = \{2\}$, $\pi = 2$, and $u = 1$. Pull the bridge back along $z \mapsto z^3$ and put $A = 1 + 4a^2$, $B = 2b$, $\delta_\tau = 1 - A\tau^2$, and $c = h(a, b, z^3) = a^2 z^6 g(a, b) / (1 - z^3 - a^2 z^6)$. The original block is

$$\Psi_\tau(a, b, c) : \exists y, r, s [\delta_\tau(c^2 - Ay^2) - 16B(r^2 - As^2) = 16].$$

The architectural step is to identify its witness s with the congruence parameter by imposing $a = 1 + 2s$. The resulting two-witness block is

$$\begin{aligned} \Theta_*(a, b, c, s) : \quad & \exists y, r \quad [c^2 - Ay^2 - 16Br^2 = 16 - 16ABs^2] \\ & \vee [c^2 - Ay^2 - 16ABr^2 = 16A - 16A^2Bs^2]. \end{aligned}$$

Its branches are respectively $\tau = 0$ and $\tau = 2a/A$; in the latter case $\delta_\tau = 1/A$ and the displayed equation has been multiplied by A . The candidate model is therefore

$$F(z) := \exists b, s [(1 + 2s, b) \in \Phi_1^{\{2\}} \wedge \Theta_*(1 + 2s, b, h(1 + 2s, b, z^3), s)].$$

Here $\Phi_1^{\{2\}}$ enforces $v_2(s) \geq 0$ and $b \in \mathcal{O}_2^\times$. Membership in it has existential rank at most 3; the tied block has rank 2 in (y, r) . The intersection theorem of [6] replaces their conjunction by $3 + 2 - 1 = 4$ witnesses, and projection of the two base coordinates (b, s) gives

$$2 + (3 + 2 - 1) = 6.$$

If an input set has rank zero, the elementary conjunction bound is no larger. The two branches share (y, r) and are combined by multiplying their equations; denominator clearing introduces no inverse witness because $A \equiv 5 \pmod{8}$ at 2 and $1 - z^3 - a^2 z^6$ cannot vanish over $\mathbb{Q}$.

For use by the later local model, write $Z = z^3$, $D_z = 1 - z^3 - a^2 z^6$, and $N_g(b) = 16a^4 b^2 - A(b - 1)^4$. The denominator-cleared value polynomial is

$$P(b) = 16D_z^2 A^2 b^4 - \delta_\tau a^4 Z^4 N_g(b)^2 - 32A^3 s^2 D_z^2 b^5.$$

Finally define the tied target value

$$M_\tau = 16 - \delta_\tau c^2 - 16ABs^2.$$

PROVED: soundness. CONDITIONAL: completeness. Soundness, $F(z) \Rightarrow z \in \bigcup_{w \notin S} \mathfrak{m}_w$, is proved; the converse is the open assembly lemma. If that lemma holds, the union outside S is $\exists_6$; adjoining the finitely many omitted maximal ideals preserves that count, so $\mathbb{Q} \setminus \mathbb{Z}$ is $\exists_6$, $\mathbb{Z}$ is $\forall_6$, and $\text{efd}_{\mathbb{Q}}(\mathbb{Q} \setminus \mathbb{Z}) \leq 5$.

2.3. Local structure and the exact global obstruction. PROVED (W0).

The first criterion, W0, is uniform even in the residue field $\mathbb{F}_3$. For every finite field k of odd order,

$$\sum_{a \in k} \chi(1 + 4a^2) = -1.$$

Hence some a makes A a nonsquare, and the canonical selector is

$$\tau_*(a) = \begin{cases} 0, & \chi(-1) = -1, \\ 2a/A, & \chi(-1) = 1. \end{cases}$$

It satisfies $\delta_{\tau_*} \neq 0$ and $\chi(-\delta_{\tau_*} A) = 1$: the two values of $-\delta_{\tau_*} A$ are $-A$ and -1 . Since $s \mapsto 1 + 2s$ is a bijection, tying loses no residue class and removes L5's $\mathbb{F}_3$ vacancy.

PROVED (W1).

For W1, let w be odd with $v_w(A) = v_w(\delta_\tau) = 0$, $v_w(b) = 1$, $v_w(z) \geq 1$, and $v_w(s) \geq 0$. The cube pullback gives either $c = 0$ or

$$v_w(c) = 2v_w(a) + 6v_w(z) - 2 \geq 6v_w(z) - 2 \geq 4.$$

Then the tied conic

$$-\delta_\tau A y^2 - 16B r^2 = 16 - \delta_\tau c^2 - 16ABs^2$$

is soluble over $\mathbb{Q}_w$ if and only if $\chi_w(-\bar{\delta}_\tau \bar{A}) = 1$. The right side is a unit congruent to 16; its two subtracted terms have valuations at least 8 and 1. Thus the target-place test is independent of the tied value of s .

PROVED (W2).

W2 identifies the remaining obstruction. Put $E_\tau = \mathbb{Q}(\sqrt{-\delta_\tau AB})$. The conic is equivalent to

$$N_{E_\tau/\mathbb{Q}}(\delta_\tau A y + 4r\sqrt{-\delta_\tau AB}) = -\delta_\tau A M_\tau, \quad M_\tau = 16 - \delta_\tau c^2 - 16ABs^2.$$

Its global solubility is exactly

$$M_\tau = 0 \quad \text{or} \quad -\delta_\tau A M_\tau \in N_{E_\tau/\mathbb{Q}}(E_\tau^\times).$$

The zero alternative is genuine. In the two branches the data are

τ	E_τ	M_τ
0	$\mathbb{Q}(\sqrt{-AB})$	$16 - c^2 - 16ABs^2$
$2a/A$	$\mathbb{Q}(\sqrt{-B})$	$16 - c^2/A - 16ABs^2$.

2.4. Structural corollaries and the remaining lemma. PROVED (L7a). At an odd place with $v(\delta_\tau A) = v(B) = 0$ and $M_\tau \neq 0$, the tied conic is soluble when $v(M_\tau)$ is even; when that valuation is odd, it is soluble exactly when $\chi_v(-\delta_\tau AB) = 1$. If also $v(A) = 0$, the corresponding untied ternary form is universal over $\mathbb{Q}_v$. Accordingly the full candidate obstruction set is

$$\{2, \infty\} \cup \{v : v(M_\tau) \text{ odd}\} \cup \{v : v(A) \neq 0 \text{ or } v(B) \neq 0\}.$$

Coefficient-bad places are genuine and cannot be replaced merely by the support of AB , because valuations can cancel in that product.

CONDITIONAL (L7b, conditional on L6 soundness). No fixed admissible (s, b) and branch admits rational functions $Y(z), R(z) \in \mathbb{Q}(z)$ satisfying

$$-\delta_\tau A Y(z)^2 - 16B R(z)^2 = M_\tau(z)$$

identically. This rules out a uniform rational section, not pointwise coverage of all relevant rational fibres by some fixed pair.

PROVED (L7c). For $M_\tau \neq 0$, local solubility is equivalent to

$$t_v := (-16\delta_\tau AB, -\delta_\tau A M_\tau)_v = 1, \quad \prod_v t_v = 1.$$

Thus a failing tied conic is obstructed at an even number of places, at least two, and repairs must change places in pairs.

CONDITIONAL / OPEN (assembly). For every odd target w and every z with $v_w(z) \geq 1$, one must choose (s, b) with $(1 + 2s, b) \in \Phi_1^{\{2\}}$, $s \equiv (\bar{a}_w - 1)/2 \pmod{\mathfrak{m}_w}$, $v_w(b) = 1$, and one of the two branches such that $M_\tau = 0$ or $-\delta_\tau A M_\tau \in N_{E_\tau/\mathbb{Q}}(E_\tau^\times)$. Equivalently, the free b -direction must steer the quadratic field $E_\tau = \mathbb{Q}(\sqrt{-\delta_\tau AB})$ simultaneously against the moving odd-valuation places of M_τ and the moving coefficient-bad places. That weak/norm-approximation statement, not W0, W1, W2, or the quantifier bookkeeping, is the open problem.

EVIDENCE. Bounded witness-switching searches support this steering mechanism, but are not a proof of assembly and are not used to upgrade the conditional six-quantifier conclusion.

3. OBSTRUCTION WALLS AND RECIPROCITY STEERING

3.1. The alignment wall. Put

$$a = 1 + 2s, \quad A = 1 + 4a^2, \quad B = 2b, \quad P = 1 - ABs^2.$$

For the two original tied branches, the source square classes are $D_0 = P$ and $D_1 = AP$, whereas the target norm fields are $E_0 = \mathbb{Q}(\sqrt{-2Ab})$ and $E_1 = \mathbb{Q}(\sqrt{-2b})$. The exact value identities show that the branch value u_τ is a norm from $\mathbb{Q}(\sqrt{D_\tau})$, while solubility asks that Au_τ be a norm from E_τ . In both branches the mismatch is measured by

$$D_\tau \text{ disc}(E_\tau) \equiv -2AbP =: \kappa \pmod{\mathbb{Q}^{\times 2}}.$$

Proposition 3.1 (L8: absorption and the parity wall). **PROVED.** *At an odd place v , assume the branch value M_τ is nonzero and each of $v(\delta_\tau A)$, $v(B)$, and $v(M_\tau)$ is even. Then the tied conic has a $\mathbb{Q}_v$ -point. Consequently an odd obstruction can occur only where A , B , or M_τ has odd valuation. On the other hand, for every admissible pair, $v_2(P) = 0$ and $v_2(\kappa) = 1$.*

Proof. After scaling the two conic variables and dividing by the even power of v contributed by M_τ , one obtains a nondegenerate binary quadratic form with unit coefficients representing a unit. Over the residue field it represents every nonzero class: in the anisotropic case it is the norm form of the quadratic extension, whose norm is surjective. Hensel lifting gives the local point. This is the absorption mechanism.

Admissibility gives $v_2(ABs^2) \geq 1$, hence P is a 2-adic unit. Thus $-2AbP$ has odd 2-adic valuation. The target discriminants $-2b$ and $-2Ab$ likewise have valuation one, so the target is always a field and can never equal the source quadratic field. $\square$

The two exact-matching devices are therefore empty on the admissible locus. Indeed, matching source and target would require $\text{disc}(E_\tau) \equiv D_\tau$, equivalently $\kappa \equiv 1$ modulo squares; splitting the target would require $\text{disc}(E_\tau)$ itself to be a square. The first condition contradicts $v_2(\kappa) = 1$, and the second contradicts the odd valuation of the target discriminant. This rules out precisely these two identities, not every possible fixed square-class relation: for a general class j , one would still have to prove that an odd-multiplicity value prime inert in $\mathbb{Q}(\sqrt{j})$ must occur, and no such theorem is asserted here.

EVIDENCE. The frozen rescues do not exhibit a further support-controlled identity: all ten have odd-multiplicity wild primes, and all measured symbols at those primes are $+1$. In the recorded session sweep, thirteen of 161 admissible pairs succeeded, while every observed obstruction set had even cardinality. These observations are alignment statistics only; they do not prove a uniform mechanism or an assembly theorem.

3.2. Reciprocity steering.

Lemma 3.2 (L9 steering lemma). **PROVED.** *For $x, d \in \mathbb{Q}^\times$, let*

$$T = \{2, \infty\} \cup \{q \text{ odd} : v_q(x) \neq 0 \text{ or } v_q(d) \neq 0\}.$$

Then $\prod_{v \in T} (x, d)_v = 1$. Hence, if all symbols in T except possibly one are $+1$, the remaining symbol is also $+1$.

Proof. Hilbert reciprocity gives the product over all places. Outside T , both entries are odd-adic units and their Hilbert symbol is $+1$, so only the displayed finite set remains. $\square$

For the tied conic take $x = \alpha M_\tau$, $d = \alpha B$, and $\alpha = -\delta_\tau A$. Once every controlled symbol is $+1$, a single wild odd-multiplicity prime q_0 with $v_{q_0}(d) = 0$ is not an obstruction: its symbol is forced by the lemma. Thus the useful arithmetic target is not complete smoothness, but a controlled part together with one proven prime-power cofactor.

Proposition 3.3 (L9a: the prime- b symbol). **PROVED.** Fix an admissible a , a cell (w, z) , and a branch τ . Let $b = \varepsilon q_1$, where q_1 is odd prime, and impose separately

$$v_{q_1}(\delta_\tau) = v_{q_1}(A) = v_{q_1}(a) = v_{q_1}(z) = v_{q_1}(D_z) = 0.$$

Then

$$(x, d)_{q_1} = \left(\frac{A}{q_1} \right).$$

In particular, the moving place is aligned by choosing a residue class with $\left(\frac{A}{q_1} \right) = +1$.

Proof. Clearing denominators and reducing at $b \equiv 0 \pmod{q_1}$ gives $N_g(0) = -A$. The hypotheses make this a unit, the denominator contributes valuation four, and $v_{q_1}(x) = -4$. Its unit part is A modulo squares, while $v_{q_1}(d) = 1$; the Hilbert-symbol parity formula gives the stated Legendre symbol. Quadratic reciprocity makes it constant on suitable congruence classes. $\square$

The corrected progression freezes only genuine controlled places. With $S_0 = \{2\} \cup \text{supp}(\alpha)$, the modulus contains the required 2-adic and A -character moduli and the Taylor exponents at places of S_0 ; it does not freeze the moving prime. For a coprime prime q_1 in the selected class,

$$\text{supp}(d) = S_0 \cup \{q_1\}, \quad \left(\frac{A}{q_1} \right) = +1.$$

Thus every place in the support of d is controlled, and only the odd-multiplicity primes of the remaining cofactor of x are wild.

The need for the corrected modulus is witnessed by the recorded cell $(w, u) = (73, 5)$. With $s = 0$, $b_0 = -29$, $\tau = 2/5$, the superseded recipe gives $N = 689120$. The prime $q = 244637629 = 29 + 355N$ has the same prescribed class, sign, and value of $(A \mid q)$, but at 59 the symbol changes: $v_{59}(x(b_0)) = 0$ with symbol $+1$, whereas $v_{59}(x(-q)) = 1$ with symbol -1 .

CONDITIONAL. Turning such a progression into an infinite assembly family still requires the named Schinzel-H/Bunyakovsky input that simultaneously makes q_1 prime and leaves a single prime cofactor of x . Reciprocity removes the last symbol once that input is supplied; it does not prove the prime-value assertion.

3.3. The generalized coprimality wall.

Theorem 3.4 (L12a). **PROVED.** Let a be admissible and suppose $A = 1 + 4a^2$ is prime. Let τ be any branch, assume $v_A(z) \leq 0$, and suppose every odd-valuation place of an admissible rational b is coprime to $\{2, z, D_z, a, \delta_\tau, A\}$. If b_{sf} denotes the squarefree part of its odd part, then

$$\prod_{p \in \{A\} \cup \text{oddsupp}(b)} (x, d)_p = -1.$$

Thus at least one of these places obstructs every branch.

Proof sketch. At each odd place p of b , whether p occurs in the numerator or denominator, the dominant-term calculation makes $v_p(x)$ even and its unit part

$$u_x \equiv A \pmod{\mathbb{Q}_p^{\times 2}}.$$

Therefore $(x, d)_p = (A \mid p) = (p \mid A)$, and multiplication over these places gives $(b_{\text{sf}} \mid A)$. At A , the branch-uniform parity calculation gives $(x, d)_A = -(b_{\text{sf}} \mid A)$: the minus sign is $(2 \mid A) = -1$, since $A \equiv 5 \pmod{8}$, while the branch-dependent unit of α is a residue. The two Jacobi factors cancel and leave -1 . $\square$

The only door in this argument is non-coprimality. If b shares a prime f with the cell data, the reduction to the constant term at f and hence $u_x \equiv A$ can fail; the telescoping proof then no longer applies. This is a door, not by itself a solubility claim.

3.4. Composite squarefree wall.

Theorem 3.5 (L13e). ***PROVED.** Let a be admissible, put $A = 1 + 4a^2$, and let K be the squarefree kernel of its rational square class; this includes composite, non-squarefree, and rational A . Assume $v_p(z) \leq 0$ for every odd $p \mid K$ and that every odd-valuation place of an admissible rational b is coprime to $\{2, z, D_z, a, \delta_\tau, A\}$. Then, for every odd $p \mid K$ and every odd ℓ occurring to odd order in b ,*

$$(x, d)_p = \left(\frac{2b}{p}\right), \quad (x, d)_\ell = \left(\frac{A}{\ell}\right),$$

and

$$\prod_{p \mid K} (x, d)_p \prod_{\ell \in \text{oddsupp}(b)} (x, d)_\ell = \left(\frac{2}{K}\right)^{1+v_2(b)}.$$

Proof sketch. At an odd $p \mid K$, the middle term $-\delta_\tau a^4 Z^4 N_g^2$ strictly dominates the other terms of the value polynomial, and $v_p(N_g) = 0$. Hence $v_p(x)$ is odd and its unit class contains the complementary factor K/p . Every prime divisor of K is $1 \pmod{4}$, because it divides a value $n^2 + 4m^2$ with the data coprime. In the Hilbert-symbol formula, the complementary factors from the two entries cancel. This yields $(2b \mid p)$ independently of the branch.

Multiplying over $p \mid K$ and over the moving primes of b produces the cross terms $(M \mid K)(K \mid M)$. Quadratic reciprocity cancels them factor by factor because every divisor of K is $1 \pmod{4}$. Moreover $K \equiv A \equiv 5 \pmod{8}$, so $(2 \mid K) = -1$. Admissibility is exactly $v_2(b) = 0$, for which the product is -1 ; at the first non-admissible boundary $v_2(b) = 1$, the product is $+1$ and this wall dissolves. Even-exponent primes of A do not enter K . $\square$

It follows that the necessity direction of L11h is a theorem for every admissible a : escaping the coprime wall requires a prime $p \mid K$ with $v_p(z) > 0$, and every such p is congruent to 1 modulo 4 and divides the numerator of z . Hence reachability implies that the numerator of z has a prime divisor congruent to 1 modulo 4. This conclusion does not supply the separate prime-value input needed for assembly.

4. CLASS DICHOTOMY, CHARACTERIZATION, AND ESCAPES

Fix an admissible parameter a , put

$$A = 1 + 4a^2, \quad \delta_\tau = 1 - A\tau^2, \quad \alpha = -\delta_\tau A,$$

and first consider the prime-member family $b = \varepsilon q$, with $\varepsilon \in \{\pm 1\}$ and q prime. An *aligned class* means a nonempty prime progression in which the Hilbert symbols at every controlled finite place, at the moving place q , and at the real place are all $+1$. This is deliberately a frozen-side notion: odd-valuation primes emerging from the value of $P(b)$ are a separate obligation.

4.1. The L10 dichotomy.

Proposition 4.1 (Canonical alignment and the opposite wall; PROVED in the stated scope). *On the canonical branch $\tau = 2a/A$, the prime- b family is aligned with the canonical square class -1 in the exact sense that*

$$\delta_\tau = \frac{1}{A}, \quad \alpha = -1.$$

On the branch $\tau = 0$, if a is odd, A is prime, $A \nmid 2zD_z$, and the prime $q \neq A$ satisfies the prescribed unit conditions, then no aligned class exists.

The first assertion is an identity, not permission to discard the places above A : although α is an A -unit, $\delta_\tau = 1/A$ forces those places back into the controlled set. This distinction is essential; the

obsolete grid count obtained by omitting them is not used here. For the second assertion, $\tau = 0$ gives $\delta_\tau = 1$ and $\alpha = -A$. The local calculation at A yields

$$(x, d)_A \left(\frac{A}{q} \right) = \left(\frac{-2\varepsilon}{A} \right) = -1.$$

Indeed admissibility gives $A \equiv 5 \pmod{8}$, so $\left(\frac{-1}{A}\right) = +1$ and $\left(\frac{2}{A}\right) = -1$, independently of the sign ε . Two required symbols are therefore permanently anti-correlated. This is a theorem for prime integral A under the displayed hypotheses; it is not a claim extracted from a finite failure search.

4.2. Free branches, an immovable wall. Branch completion itself costs nothing. If $\tau = p/q$ is rational in the existing parameters and its denominator is proved nonzero on the base locus, adjoining that branch introduces no inverse witness. Admissibility makes A a nonsquare in $\mathbb{Q}_2$, hence $\delta_\tau \neq 0$; the norm identity in the unreferenced argument of Sun [13] is uniform in τ ; and a finite disjunction of branch equations is represented by their product in the same witnesses. Explicit choices meet all four square classes of $\langle -1, A \rangle$, including the square branch

$$\tau^\dagger = \frac{1 + 2a^2}{1 + 4a^2}, \quad \alpha = (2a^2)^2.$$

The corrected controlled set for a prime member is

$$S = \{2, 3, 5, 7\} \cup \text{supp}(\alpha) \cup \text{supp}(\delta_\tau).$$

Outside S , the fixed content of P has even valuation; after that content is removed, the reduction is a nonzero polynomial whose degree prevents a permanent odd valuation at the remaining primes. Thus no fixed bad place is silently omitted.

More importantly, changing branches does not change the obstruction. For prime A , admissible a , and $v_A(z) \leq 0$, every branch in the relevant square-class group satisfies

$$(x, d)_A(x, d)_q = -1.$$

Hence at least one controlled symbol is negative. This branch-independent wall was checked exactly on 2,136 extended instances across all four explicit branches; the computation confirms the proved identity rather than replacing it.

Theorem 4.2 (The L11h characterization; PROVED). *For the prime-b family, an aligned class exists at the cell (w, u) , with $z = wu$, if and only if the numerator of z has a prime divisor $p \equiv 1 \pmod{4}$. On the finite grid this selects 190 of 353 cells; the other 163 cells are unreachable by this family on every branch.*

Proof sketch. For necessity, the generalized wall says that alignment is impossible unless some odd prime $p \mid A$ also has $v_p(z) > 0$. Write $a = m/n$ in lowest terms. Such a prime divides $n^2 + 4m^2$, and therefore $(2mn^{-1})^2 \equiv -1 \pmod{p}$; consequently $p \equiv 1 \pmod{4}$. The factor-by-factor telescoping argument applies also when A is composite or rational, so this necessity is PROVED for every admissible a , not merely supported by a search.

Conversely, let $p \equiv 1 \pmod{4}$ divide the numerator of z . Choose odd representatives m, n satisfying $n^2 \equiv -4m^2 \pmod{p}$ and set $a = m/n$. Then a is admissible and $v_p(A) > 0$, so the factor shared by A and z breaks the wall. Freezing the complete controlled set gives an explicit aligned progression. The construction, including its escape prime, chosen a , branch, and modulus, is replayed on all 190 predicted rows. This constructive sufficiency and the preceding necessity make the equivalence the structural centre of the class analysis. $\square$

4.3. Non-coprime escapes and complete grid coverage. The characterization concerns $b = \varepsilon q$. The only effective door through the generalized coprime wall is to share a cell prime deliberately:

$$b = \varepsilon f q, \quad f \mid z.$$

This produces aligned escape classes at 103 of the 163 prime-family-walled cells. Their frozen-side alignment is PROVED row by row; the remaining 60 walled cells with no such escape are closed instead by direct, fully soluble witnesses.

The Schinzel audit also keeps six legacy residual rational/composite- b witnesses separate as pointwise certificates. They are unconditional local witnesses, but they are not promoted into prime progressions and give no prime-value theorem. The eventual last cell, $(89, -2)$, was closed by the replayed square-branch witness $(a, b, \tau) = (3, 89/367, 19/37)$, completing cell coverage.

verified object	coverage	status
L11 prime- b aligned classes	190/190	PROVED PER ROW
non-coprime escape classes	103/103	PROVED PER ROW
all aligned classes, with a zero-bad member	293/293	PROVED PER ROW
all grid cells, with a soluble witness	353/353	PROVED PER ROW

The class total is exactly the disjoint sum of the L11 and escape families; the cell total additionally uses direct pointwise witnesses. Neither total asserts a uniform theorem beyond the verified grid.

4.4. Why no square-class shortcut remains.

Proposition 4.3 (L13c; PROVED per row). *For every one of the 706 cell-branch rows, P is irreducible of degree 8 over $\mathbb{Q}$. Consequently no constant-square or weak square-factor form collapses the emergent-place obligation on the grid.*

The boundary between theorem and experiment is sharp. The irreducibility certificates are PROVED. By contrast, the 137.5 million exact square-class tests with no admissible hit are EVIDENCE against a hidden special-value shortcut, not a universal nonexistence theorem. Likewise the observed Galois group $D_8 \wr C_2$ is EVIDENCE: its cycle types matched all 28,240 certified unramified Frobenius samples, but no group-identification proof is claimed.

Finally, the earlier zero-bad search already supplied 24 certified members across 15 distinct cells; those are PROVED INSTANCES, while the rough quarter-density inferred from the sample is only EVIDENCE. The later per-class replay strengthens the finite conclusion to the 293/293 line in the table, without changing the open uniform problem.

5. THE VERIFIED QUANTITATIVE LAYER

The class construction reduces the residual local problem to a moving arithmetic progression. For an aligned class write

$$Q(k) = q_1 + kN, \quad b(k) = \varepsilon f Q(k),$$

with the controlled Hilbert symbols already fixed to be positive. A member is called *emergent-free* when the remaining odd-valuation places contribute no negative symbol. The purpose of the quantitative layer is twofold: to certify this condition on the finite grid and to separate the exact local mechanism from the statistical model suggested by the resulting population.

5.1. L14: instance verification.

Proposition 5.1 (Finite instance of hypothesis H). **PROVED (per row).** *All 293/293 aligned classes in the finite grid have a certified emergent-free member; these comprise 190 L11 classes and 103 escape classes. Every stored row was replayed through the exact cofactor ladder and returned verdict zero.*

The proposition is deliberately finite. Its acceptance condition is an aligned class together with an exact zero verdict from the member ladder. A tied-conic or ramification computation may also be attempted as an auxiliary cross-check, but a budget refusal there is accepted and logged: it neither invalidates a ladder certificate nor supplies positive evidence. In particular, no refusal is promoted to a primality or solubility assertion. This distinction is essential because the general, all-cell form of hypothesis H remains open even though the displayed grid instance is complete.

5.2. L15: the exact remainder and counting laws. For a certified member, let R be the remainder of $P(b)$ after the frozen support, the fixed shell, and the smooth-emergent primes have been stripped. An independent replay gives the following exhaustive shape table.

Ladder shape	Rows	Exact form of R	Size data
proved-prime	197	one prime, never pC^2	median 48 digits; maximum 119
factorint	96	squarefree product, $e \in \{2, 3, 4\}$	multiplicities 76/15/5

TABLE 1. **PROVED (finite audit).** Shapes of the stripped remainder on all 293 closure rows.

Thus R is squarefree on every audited row and $R = 1$ never occurs. The prime rung is not a probable-prime shortcut: its singleton cofactor is accepted only after a proof of primality. On a factorint rung the integer e counts distinct emergent factors, and the exact factorization allows every corresponding Hilbert symbol to be checked. The two shapes therefore reflect two proof routes through the same local criterion, not two definitions of closure.

There is also an exact finite counting statement. The first certified member occurs at $k = 0$ in 207/293 classes, or 70.65%. This is **PROVED** as a statement about deterministic playback of the stored progressions. By contrast, interpreting the searches as samples from a stable member process is **EVIDENCE**: the measured per-prime-member emergent-free rate is approximately 0.040, and the observed waiting time is approximately 15.7 times the pure Bateman–Horn [1] first-prime prediction for the same progressions. The exact histogram may therefore be used without qualification inside the finite audit, while the rate and the comparison are empirical laws, not uniform density theorems.

5.3. L16: why the small layer has sieve shape. Fix an emergent odd prime p . On an aligned class, $b(k)$ is affine in k , and the sign at p is

$$(x_0, d_0)_p = \left(\frac{2\alpha b(k)}{p} \right).$$

On every checked recurring-prime row, this sign is determined by $k \bmod p$. The obstruction at p is consequently not the whole root set of $P(b(k))$ modulo p : it is the union of those root classes having the bad signature, further restricted to odd valuation,

$$\mathcal{B}_p = \{k : v_p(P(b(k))) \text{ is odd}\} \cap \{k \bmod p : \left(\frac{2\alpha b(k)}{p} \right) = -1\}.$$

For a simple bad root, Hensel lifting gives an exact Haar mass

$$\sum_{j \geq 0} \frac{p-1}{p^{2j+2}} = \frac{1}{p+1}.$$

This is the valuation-parity correction to a naive exclusion sieve: one must retain the odd levels of the Hensel tower rather than merely delete a residue class modulo p . A singular root has no such one-line factor. Its descendants are lifted recursively until the odd and even masses are certified with a residual bound; if that cannot be completed, the corresponding local product is explicitly labeled

as an exclusion or partial result. Accordingly, the residue and valuation checks are **PROVED per row**; the claim that this small layer supplies the useful global layering model is **EVIDENCE**. No independence theorem is hidden in the product notation.

5.4. L17: matched validation. The full-grid residue test strengthens the preliminary structural probe. Across all 293 classes and the window $k \in [0, 119]$, it found zero residue-determination counterexamples on the identical 8,760-member sample used for evaluation. This is **PROVED** as a finite statement. The predictive use of the local masses is **EVIDENCE**; it was assessed only after the predictions and observations had been put on exactly the same member support.

Matched statistic	Model	Observed	Diagnostic
marginal odd hits	4481.3	4478	ratio 0.9993
hit-conditioned union	3477.9	3475	residual -2.9
permutation null	—	3475	$z = -0.09$; tails 0.61/0.53

TABLE 2. **EVIDENCE**. Matched small-layer evaluation on the same 8,760 members used on both sides.

The marginal statistic counts odd bad-root hits, so a member can contribute more than once; the union statistic asks whether a member is hit at least once. Their simultaneous agreement, together with the marginal-preserving permutation null, shows no detectable overlap or dependence discrepancy on this window. It does not prove independence outside the evaluated support.

A superficially different class-weighted comparison has estimand about $+0.0195$, at 3.63 standard deviations. That quantity compares Haar-uniform local masses in $\mathbb{Z}_p$ with the finite $k \in [0, 119]$ window, rather than comparing predictions and observations member by member on matched support. It is therefore identified as the Haar-window gap, not as a failure of the hit-conditioned model.

5.5. L17 cofactor layer: reciprocity and its limit. The large-cofactor layer has a narrower theorem than the small layer. **PROVED:** on the prime rungs, reciprocity forces the singleton cofactor sign to be $+1$ in all 197/197 cases; on the complete closure set, the cofactor-sign product is $+1$ on 293/293 rows. These are exactly the singleton and parity-product statements. The individual positive signs in factorint closure rows are selected by the closure condition, so their residue tables are descriptive data and cannot support an unconditional sign-distribution claim.

The attempted nonclosure measurement remains **INCONCLUSIVE**: only 11/400 sampled rows were resolved, and the predeclared power gate failed. The unresolved rows are refusals, not observations of either sign. Consequently the cofactor layer is proved at reciprocity level and no further; its nonclosure sign distribution remains measurement-limited. This is also why the matched small-layer success must not be silently upgraded into a complete rate theorem for emergent-free members.

6. THE CONDITIONAL RECORD AND THE SCHINZEL AUDIT

6.1. The statement, with its hypothesis exposed. The hypothesis used below is not an unnamed appeal to genericity. It is the following cellwise assertion, quoted in the form in which it is recorded in the ledger.

Definition 6.1 (CONDITIONAL: Hypothesis H). For every odd prime w and every rational z with $v_w(z) \geq 1$ there exists a verified aligned class for the cell (w, z) —prime- b data $(a, \varepsilon, 1, q, N)$ from the L11 family, or escape data $(a, \varepsilon, f, q, N)$ with $f \mid \text{num}(z)$ from the L12 family—and a prime member $Q \equiv q \pmod{N}$ outside its finite excluded set, at which the emergent set is empty: every odd prime $l \notin S \cup \{Q\}$ satisfying $v_l(P(\varepsilon f Q)) \neq 0$ has $v_l(P(\varepsilon f Q))$ even. Here S is the frozen controlled set belonging to the verified class.

Theorem 6.2 (CONDITIONAL: Theorem C). *Assume H. Then $\mathbb{Q} \setminus \mathbb{Z}$ is Diophantine in $\mathbb{Q}$ with six unknowns, realized by the L6 formula*

$$F(z) := \exists b, s \left[(1 + 2s, b) \in \Phi_1^{\{2\}} \wedge \Theta_*(1 + 2s, b, h(1 + 2s, b, z^3), s) \right],$$

where, with $A = 1 + 4a^2$ and $B = 2b$,

$$\begin{aligned} \Theta_*(a, b, c, s) := \exists y, r \quad & [c^2 - Ay^2 - 16Br^2 = 16 - 16ABs^2] \\ & \vee [c^2 - Ay^2 - 16ABr^2 = 16A - 16A^2Bs^2]. \end{aligned}$$

Consequently $\mathbb{Z}$ is definable in $\mathbb{Q}$ by a formula with six universal quantifiers, $\mathbb{Z} = \mathbb{Q} \setminus (F \vee \mathfrak{m}_2)$, and $\text{efd}_{\mathbb{Q}}(\mathbb{Q} \setminus \mathbb{Z}) \leq 5$.

This is the ledger's exact conditional conclusion. It is *not* an unconditional definition and makes **no record claim**. Soundness of the displayed formula consumes no hypothesis; completeness is exactly H. A finite collection of closed cells, however exhaustive within its stated horizon, cannot replace H's quantification over every odd w and every eligible z .

The quantifier saving is attributed precisely to Nicolas Daans, Philip Dittmann, and Arno Fehm, *Existential rank and essential dimension of diophantine sets*, arXiv:2102.06941v5, Theorem 1.4 (equivalently Corollary 5.11) [6]. Their intersection theorem turns the ranks three and two on the common base into $3 + 2 - 1 = 4$ witnesses; the two base coordinates then give six unknowns. The recorded rank–essential-dimension identity gives the bound five. These counts remain **CONDITIONAL** because the set equality to $\mathbb{Q} \setminus \mathbb{Z}$ uses H.

6.2. What is proved, and what Schinzel must supply. PROVED (finite alignment settled). Once an aligned class has been found, all controlled Hilbert symbols are frozen and the moving prime is restricted to one reduced progression. For every canonical row audited here this progression is

$$q_1(t) = q_{1,0} + Nt, \quad \gcd(q_{1,0}, N) = 1.$$

The stored $q_{1,0}$ is proven prime by the repository's proven-primality engine. Dirichlet's theorem therefore supplies infinitely many prime values of $q_1(t)$. This is the linear, unconditional half of the member-existence problem; neither a refusal by the primality engine nor a probable-prime test is used as evidence.

OPEN/CONDITIONAL (cofactor side). Put

$$F_{\text{cell}}(t) = P(\varepsilon f q_1(t)),$$

using the primitive integer representative after removal of the fixed rational scale recorded in the audit. Requiring $F_{\text{cell}}(t)$ to be prime simultaneously with $q_1(t)$ is the two-polynomial instance of Schinzel's Hypothesis H [12] (or the corresponding Bunyakovsky-type prime-value input). The polynomial F_{cell} has degree eight. A prime cofactor is a sufficient route to an emergent-free member, because the class has already frozen the controlled places and reciprocity forces the remaining single symbol. This sufficient prime-cofactor route must not be confused with a proof of H, whose conclusion asks only for one emergent-free member and is formally weaker. In the ledger's shorthand, after using the progression prime itself as variable, the pair is written $\{t, F_{\text{cell}}(t)\}$.

6.3. PROVED finite audit of the Schinzel pair. The exact audit contains 293 canonical records, split into 190 L11 classes and 103 escape classes. The fields `F_irreducible`, `F_value_gcd`, and `pair_product_gcd` respectively certify irreducibility on 293/293 rows, value gcd one on 293/293 rows, and pair-product gcd one on 293/293 rows. Every irreducibility certificate is a degree-eight Frobenius certificate modulo a certified prime; both gcds were computed exactly on $0 \leq t \leq 2000$. The pair-product computation is already a certificate that no prime divides the product for every integer t : any such fixed divisor would divide the computed gcd. Thus all canonical pairs pass the

fixed-divisor and irreducibility conditions. This is **PROVED** for the stored pairs, not a prime-value theorem.

For a prime p dividing N , the corrected residue count is

$$\nu_p = \#\{t \in \mathbb{F}_p : q_1(t) \not\equiv 0 \pmod{p}, F_{\text{cell}}(t) \not\equiv 0 \pmod{p}\}.$$

The first condition is essential. Since $p \mid N$ and $\gcd(q_{1,0}, N) = 1$, the linear polynomial is the nonzero constant $q_{1,0}$ modulo p ; it must not be replaced by the parameter t . Consequently the count is p , rather than $p - 1$, whenever the cofactor has no root modulo p . The audit enumerated every $p \leq 1000$ dividing the row's modulus and obtained exactly this full count in every occurrence.

primes p	records containing $p \mid N$	range of ν_p	mean density ν_p/p
2, 3, 5, 7	293 each	p	1
11, 37, 89, 97	13 each	p	1
13, 53	26 each	p	1
17	39	p	1
23	38	p	1
29, 41, 61, 73	28 each	p	1
43, 47, 67, 83	10 each	p	1

TABLE 3. Corrected admissible-residue counts for primes dividing the progression modulus.

These local counts neither assume independence nor estimate a density of simultaneous prime values. Their role is narrower and exact: together with the pair-product gcd, they rule out the elementary local obstruction for each stored Schinzel pair.

6.4. Residual certificates and the literature boundary. The six residual rows are deliberately segregated. They are rational or composite- b witnesses verified by exact branch reconstruction and tied status; they are not coerced into the prime family and are not evidence for Schinzel's hypothesis.

cell (w, z)	(a, b)	τ	status
(67, 2)	(1, 4757)	3/5	PROVED pointwise
(41, 1/7)	(25, 4985)	1251/2501	PROVED pointwise
(61, -2/7)	(25, 55571)	1251/2501	PROVED pointwise
(61, 2)	(25, -71431)	1251/2501	PROVED pointwise
(53, 1/3)	(-15, 9)	451/901	PROVED pointwise
(29, -2)	(-15, -29/5)	4055/2703	PROVED pointwise

TABLE 4. The six residual witnesses: unconditional certificates for their individual cells only.

OPEN (published boundary). The exact square-class reference is David Krumm, *Squarefree parts of polynomial values*, Journal de Théorie des Nombres de Bordeaux **28** (2016), no. 3, 699–724, arXiv:1407.4890 [10]. Its Theorem 1.3 and Propositions 3.4–3.5 give the unconditional degree-one and degree-two cases. Proposition 3.8 makes the degree-three case conditional on the Parity Conjecture for elliptic curves over $\mathbb{Q}$ —rank parity, not the sieve “parity problem.”

At prime arguments, the nearest unconditional results reach the cubic boundary without prescribing a square class: Helfgott proves square-free values of $f(p)$ for separable cubic f [8], while Reuss proves $(d - 1)$ -free values rather than prime or prescribed-square-class values [11]. Neither yields the required degree-eight conclusion.

The conic-bundle literature does not remove the gap. The unconditional Harpaz–Skorobogatov–Wittenberg theorem requires abelian constant fields in the bad fibres [7, Theorem 3.1 and Corollary 3.4]. The general Schinzel reference in *J. reine angew. Math.* **495** (1998), 1–28 is by Colliot-Thélène, Skorobogatov and Swinnerton-Dyer [3], not by Colliot-Thélène and Sansuc; the latter

pair's Schinzel paper is [2]. Thus no unconditional result is available for the degree-eight prime-value side or for the required degree-eight square-class side. The six-quantifier theorem above remains **CONDITIONAL**, and no unconditional record is claimed.

7. SCHINZEL'S HYPOTHESIS H IMPLIES THE PER-CLASS HYPOTHESIS

The conditional statement of Section 6 rests on a hypothesis H about the existence of emergent-free members. We now show that H is not a bespoke conjecture: for a fixed verified aligned class it follows from Schinzel's Hypothesis H [12] applied to an explicit pair of polynomials. The implication is a theorem; only the prime values themselves remain conjectural.

7.1. The operational contract. A place is recorded as an obstruction only when its valuation is odd *and* its Hilbert symbol is -1 : the kernel appends a place under exactly that conjunction, and the ladder returns *zero* when every remaining place is proved with symbol $+1$. Emergent-freeness therefore means *no place of symbol -1* , not *no odd valuation outside the frozen set*. This distinction is load-bearing: 197 of the 293 grid closures have exactly one odd-valuation outside prime whose symbol is forced $+1$ by reciprocity. The stronger reading would not follow from prime values, since a prime value deliberately creates one odd place.

7.2. The theorem. Fix a verified aligned class with data $(a, \varepsilon, f, q_1, N)$, put $Q(t) = q_1 + Nt$, $b(t) = \varepsilon f Q(t)$, and let $S = \{2, 3, 5, 7\} \cup \text{supp}(\alpha) \cup \text{supp}(\delta) \cup \text{supp}(f)$. Write the composed class polynomial as

$$F(t) = P(\varepsilon f(q_1 + Nt)) = cG(t),$$

with $c > 0$ supported on S and $G \in \mathbb{Z}[t]$ primitive with positive leading coefficient.

Theorem 7.1 (Schinzel implies H per class; PROVED implication). *Assume the class certificate (all symbols at S , at the moving prime Q , and at infinity equal $+1$ for every member outside a finite excluded set), and assume Q and G are irreducible with positive leading coefficients whose product has no fixed prime divisor. Then Schinzel's Hypothesis H for $\{Q, G\}$ implies that the class contains infinitely many members with ladder verdict zero.*

Proof sketch. Schinzel supplies infinitely many t with $Q(t)$ and $G(t)$ simultaneously prime; the side conditions (Cauchy bound, $b \neq 0, 1$, finitely many forbidden coincidences) exclude only finitely many t . For such t , stripping S from $F(t) = cG(t)$ removes c exactly and leaves the prime $G(t)$. At the moving prime, $b \equiv 0 \pmod{Q}$ and the kernel identity gives $P(b) \equiv -\delta a^4 Z^4 A^2 \pmod{Q}$, a unit, so $v_Q(P(b)) = 0$ and $G(t) \neq Q(t)$: there is exactly one outside place $R = G(t)$ of odd valuation. Every other outside place has even valuation with $v_p(d) = 0$, hence symbol $+1$. The certificate pins the symbols at S , at Q and at infinity to $+1$. Hilbert reciprocity — the product of all local symbols is $+1$ — then forces $(x_0, d_0)_R = +1$. So no place carries -1 and the verdict is *zero*. $\square$

7.3. The hypotheses are verified, not assumed. For all 293 canonical closure pairs of the grid we verified exactly, not statistically: G is primitive of degree 8 and irreducible with a replayed Frobenius certificate; G has positive leading coefficient; $\gcd(q_1, N) = 1$; the pair product has fixed divisor 1, certified by finite differences on $t = 0, \dots, 9$ (exact for degree 9); $G(t)$ is an S -unit for every integer t ; and the frozen symbols are constant in t , covered for 283 rows by the Taylor exponent and for the remaining 10 alternate escape rows by an explicit normalized-polynomial congruence. A 24-row cross-check (12 escape, 12 L11) re-ran the proven-primality kernel end to end: every sampled member has a single proved-prime ladder residual with symbol $+1$ and verdict *zero*.

Remark 7.2 (Exact scope). Schinzel supplies members inside a class that already exists. It does not construct an aligned class for a cell; that is the finite, structural side, characterized by the class theorem and verified 293/293 on the grid. The conditional record therefore now reads: *Schinzel's Hypothesis H plus the class-existence side implies the six-quantifier universal definition.*

8. THE STEP-(II) DENSITY LAW AND THE OFF-GRID FRONTIER

This section reports the two newest components of the program: an unconditional local bound on the sieve masses together with the exact decay law it implies, and the current state of the off-grid horizon.

8.1. An unconditional local bound. Fix an aligned class with parameters (ε, f, q_1, N) and write $F(k) = P(\varepsilon f(q_1 + kN))$ for its degree-8 class polynomial, and $u(k) = 2\alpha\varepsilon f(q_1 + kN)$ for the unit whose Legendre symbol decides bad signature. For an odd prime p let m_p denote the Haar mass in $\mathbb{Z}_p$ of the set of k with $v_p(F(k))$ odd and $\left(\frac{u(k)}{p}\right) = -1$: the local mass of the emergent obstruction.

Theorem 8.1 (Local sieve bound; PROVED). *For every aligned class of the grid,*

$$m_3 = m_5 = m_7 = 0, \quad \text{and} \quad m_p \leq \frac{8}{p+1} < 1 \quad \text{for all } p \geq 11,$$

the bound holding for singular roots as well, counted with multiplicity.

The vanishing at $p \in \{3, 5, 7\}$ is exactly the frozen-alignment condition of the class certificate: those places are controlled, and their symbols are pinned to +1 by construction. For $p \geq 11$ the bound is the root count of a degree-8 polynomial: at most eight residues can be roots, each simple root contributes Haar mass $1/(p+1)$ for odd valuation, and a singular root contributes at most its multiplicity times that mass, which is again bounded by the total degree. The bound was verified independently on every stored local pair: 102,439 pairs with $p \leq 10^4$ across the 293 classes, zero violations.

Theorem 8.1 settles a natural hope in the negative, and this is the important structural point.

Corollary 8.2 (No uniform positive mass; PROVED for the current classes). *For each of the 293 classes the splitting condition governing the bad-signature event has Chebotarev exponent $c = 1/2$: the class polynomial is irreducible with a replayed degree-8 certificate, and a replayed witness shows the bad unit is a nonsquare at a simple bad root. Consequently the truncated clean product obeys*

$$\prod_{p \leq X} (1 - m_p) \asymp (\log X)^{-1/2},$$

so no cutoff-uniform positive lower bound on the clean mass exists: the small-layer clean probability decays, slowly, to zero.

The measured exponent agrees: the root-harmonic estimate gives $c \approx 0.4778$, the product estimate $c \approx 0.4801$, and the same-member sieve between 10^4 and 10^5 gives $c \approx 0.4590$ (EVIDENCE).

The consequence for hypothesis H is a sharpening, not a defeat. The number of admissible prime members up to height X in a class grows like $X/(\log X)^2$ by the Dirichlet–Bateman–Horn heuristic [1], while each carries clean probability $\asymp (\log X)^{-1/2}$; the expected count of emergent-free members therefore still diverges, at rate $X/(\log X)^{5/2}$. What Corollary 8.2 rules out is the clean shortcut — a uniform positive constant — and what it leaves is a divergence statement of exactly the shape the measured counting law already exhibits (first emergent-free member at $k = 0$ in 207 of 293 classes).

8.2. The off-grid horizon. The grid is w prime, $w < 100$. Off-grid closures now exist for

$$w \in \{101, 103, 107, 109, 113, 127, 131, 137, 139, 149, 151, 157, 163, 167, 173\}, \quad u = -1,$$

together with $[131, [5, 1]]$, each certified by an aligned class plus an exact ladder verdict *zero*, and each replayed independently. Two routes appear: the L11 route through fresh Hensel roots of $1 + 4r^2 \equiv 0 \pmod{w}$ ($w = 101, 109, 137, 149, 157$), the canonical escape route with $f = w$ ($w = 103, 107, 113, 127, 163, 167, 173$), and the non-canonical $a=7, \tau=0$ family for the cells the canonical protocol walls ($w = 131, 139, 151$). Only $w = 179$ remains open, and only for want of a

member within budget — four aligned classes were found — so no off-grid cell is currently known to be intrinsically obstructed.

Two negative or delimiting findings carry structural content.

Theorem 8.3 (The canonical 5-wall; PROVED). *On the canonical escape route ($z = -w$, $a = 1$, $\tau = 3/5$, $f = w$), write $e = v_5(D)$. Then $v_5(x_0) = -3 - 2e$ is odd and $v_5(d_0) = 0$, so*

$$\text{Hilb}_5 = -\left(\frac{w}{5}\right)\left(\frac{q_1}{5}\right), \quad \text{Hilb}_{q_1} = \left(\frac{5}{q_1}\right),$$

and both equal +1 for some admissible q_1 if and only if $\left(\frac{w}{5}\right) = -1$. Since the L11 route requires a root of $1 + 4r^2 \equiv 0 \pmod{w}$, i.e. $w \equiv 1 \pmod{4}$, the canonical protocol is obstructed exactly for

$$w \equiv 3 \pmod{4} \text{ and } \left(\frac{w}{5}\right) = +1 \iff w \equiv 11, 19 \pmod{20}.$$

The criterion was audited on all 53 primes $w \in [101, 400]$ with both character branches of q_1 covered: 26 cells have the L11 route, 13 the canonical escape route, 14 are obstructed, and there were zero identity or classification mismatches.

The wall is a statement about the canonical protocol, not about the cells, and it breaks.

Proposition 8.4 (Breaking the wall; PROVED). *The obstructed cell $[131, [-1, 1]]$ closes with the non-canonical class $a = 7$ (so $A = 197$), $\tau = 0$, $f = 131$, $\varepsilon = +1$, $q_1 = Q = 41$, $k = 0$, prime rung, tied status exact and ramified set empty. The same family closes the other two obstructed cells met so far, $w = 139$ and $w = 151$, each at $a = 7$, $\tau = 0$, $q_1 = 41$, $k = 0$.*

Two hypotheses fail, and it is worth being exact about which. The $\tau = 0$ wall assumes $b = \varepsilon q_1$ with q_1 prime; here $b = 131 \cdot 41$ is not $\pm$ a prime, while every other hypothesis of that wall does hold ($a = 7$ odd, $A = 197$ prime, $A \nmid 2zD_z$). The generalized wall assumes b coprime to the cell data; here $f = w = 131$ divides $\text{num}(z)$. This is exactly the non-coprimality door, and the shape $b = \varepsilon f Q$ with $f \mid \text{num}(z)$ is the orthodox escape shape already used in 103 grid rows. *The novelty is the $\tau = 0$ alignment alone, not a new family of b .* The row verdict is computed by the same kernel functions as every canonical row (tied status, ramified set, and the identity $M_0 = P(b)/(b^4 D_z^2 A^2)$); the class certificate is a local generalization of the escape certificate that drops two restrictions ($a \neq 1$ allowed, f composite allowed, every prime of f frozen), with the same exponent-lemma proof.

Remark 8.5 (Composite w). Composite w is not admissible in the architecture: w indexes a place, and the residue-field construction has no meaning for composite modulus — the break occurs already at the field constructor. The downstream arithmetic does generalize once w is decomposed into genuine prime places; as an illustration the pseudo-cell $[21, [-1, 1]]$ closes with the escape route at the honest prime $f = 3$ (member $k = 0$, $Q = 11$, factorint rung, tied and ramified checks exact). This is a clarification of scope, not an extension of the horizon.

9. CONCLUSION AND OUTLOOK

The program's current position is sharp. The algebraic architecture is complete and machine-checked; the search space is delimited by two theorems (the generalized wall, valid for composite squarefree A , and the class characterization); the finite verification is exhaustive on the grid and extends off-grid to $w \in \{101, 103, 107, 109, 113, 127, 137\}$; and the residual analytic content has been isolated to a single classical statement about prime values of a degree-8 polynomial in an arithmetic progression, with all of its local conditions verified.

Three directions carry the most weight.

- (1) *The step-(ii) divergence.* Theorem 8.1 and Corollary 8.2 settle the local question: the masses are bounded by $8/(p+1)$, the frozen primes contribute nothing, and the clean

product decays like $(\log X)^{-1/2}$, so no uniform positive mass exists. What remains is to convert the resulting divergence of the expected count of emergent-free members, at rate $X/(\log X)^{5/2}$, into an existence statement; that is precisely where the degree-8 prime-value input enters.

- (2) *The protocol wall, answered.* Composite w is out of scope by construction (w indexes a place). The question raised by the first obstructed cell — for which primes w does the canonical escape route collide with the L11 route at the frozen place 5? — is now answered exactly by Theorem 8.3: $w \equiv 11, 19 \pmod{20}$. It is also no longer an obstruction, since Proposition 8.4 closes every such cell met so far off the canonical protocol. No off-grid cell is currently known to be intrinsically obstructed, and the natural next question is whether the $a = 7, \tau = 0$ family closes the obstructed congruence class in general.
- (3) *Degree-8 prime values.* Unconditional prime-value theorems stop at degree three. Any unconditional progress for a single grid class would convert the conditional statement into an unconditional one for that cell, and the audit shows there is no local obstruction anywhere on the grid.

REPRODUCIBILITY

All certificates in this paper are produced by a stdlib-only kernel with a proven-primality discipline: deterministic Miller–Rabin below 3.3×10^{24} , Pocklington with the Brillhart–Lehmer–Selfridge relaxation above, and explicit refusal labels that never count as evidence. The companion paper documents the engine, the artifact schemas, the exact reproduction commands, and the precise scope of what is regenerable from checked-in scripts.

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